

Sobolev Spaces and Calculus of Variations

Theory, Examples and Proofs

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June 22, 2026

Contents

1	Sobolev Spaces	3
1.1	Examples	3
1.2	Theorems	8
2	Dirichlet Principle	15

Chapter 1

Sobolev Spaces

1.1. Examples

Example 1.1. Let $n \geq 2$ and

$$\Omega = \{x \in \mathbb{R}^n : |x| < 1\} \quad \text{and} \quad u(x) = \frac{x}{|x|}.$$

Then

$$u \in W^{1,p}(\Omega; \mathbb{R}^n)$$

for every $1 \leq p < n$.

Proof. First observe that

$$|u(x)| = \left| \frac{x}{|x|} \right| = 1 \text{ for every } x \neq 0.$$

Hence

$$\int_{\Omega} |u(x)|^p dx = \int_{\Omega} 1 dx = |\Omega| < \infty.$$

$$\implies u \in L^p(\Omega; \mathbb{R}^n) \quad \text{for every } 1 \leq p \leq \infty.$$

Write

$$u = (u_1, \dots, u_n), \quad u_i(x) = \frac{x_i}{|x|}.$$

For $x \neq 0$ and $1 \leq i, j \leq n$, we compute

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} (x_i |x|^{-1}).$$

Using the product rule, we obtain

$$\frac{\partial u_i}{\partial x_j} = \delta_{ij} |x|^{-1} + x_i \frac{\partial}{\partial x_j} (|x|^{-1}).$$

Since

$$\frac{\partial}{\partial x_j} (|x|^{-1}) = -\frac{x_j}{|x|^3},$$

it follows that

$$\begin{aligned} \left| \frac{\partial u_i}{\partial x_j}(x) \right| &= \left| \frac{\delta_{ij}}{|x|} - \frac{x_i x_j}{|x|^3} \right| \\ &\leq \frac{1}{|x|} + \frac{|x_i x_j|}{|x|^3} \\ &\leq \frac{1}{|x|} + \frac{|x|^2}{|x|^3} \\ &= \frac{2}{|x|}. \end{aligned}$$

and hence

$$|\nabla u(x)| \leq \frac{C}{|x|}$$

for some constant $C = C(n) > 0$.

Therefore

$$\int_{\Omega} |\nabla u(x)|^p dx \leq C \int_{B_1(0)} |x|^{-p} dx.$$

The integral converges if and only if

$$p < n. \quad (\text{clarify!})$$

Hence

$$\nabla u \in L^p(\Omega; \mathbb{R}^{n \times n})$$

for every $1 \leq p < n$.

Since u is smooth on $\Omega \setminus \{0\}$ and the point $\{0\}$ has Lebesgue measure zero, while both u and its classical derivatives belong to $L^p(\Omega)$, the classical derivatives agree with the weak derivatives. Therefore

$$u \in W^{1,p}(\Omega; \mathbb{R}^n) \quad \text{for every } 1 \leq p < n.$$

□

Example 1.2. Let $0 < s < \frac{1}{2}$,

$$\Omega = \left\{ x = (x_1, x_2) \in \mathbb{R}^2 : |x| < \frac{1}{2} \right\} \quad \text{and} \quad \psi(x) = |\log(|x|)|^s.$$

We have

$$\psi \in W^{1,2}(\Omega), \quad \psi \in L^p(\Omega) \text{ for every } 1 \leq p < \infty,$$

but

$$\psi \notin L^\infty(\Omega).$$

Proof. We first show that $\psi \in L^p(\Omega)$ for every $1 \leq p < \infty$.

Since

$$\psi(x) = |\log |x||^s,$$

the only possible singularity is at $x = 0$. Moving to polar coordinates,

$$\int_{\Omega} |\psi(x)|^p dx = 2\pi \int_0^{1/2} |\log r|^{sp} r dr. \text{ (independent of } \theta \text{)}$$

Making the change of variables

$$t = -\log r, \quad r = e^{-t}, \quad dr = -e^{-t} dt,$$

we obtain

$$2\pi \int_{\log 2}^{\infty} t^{sp} e^{-2t} dt.$$

Since the exponential factor e^{-2t} dominates every polynomial growth, the above integral converges. Hence

$$\psi \in L^p(\Omega) \quad \text{for every } 1 \leq p < \infty.$$

Next we show that $\psi \notin L^\infty(\Omega)$. Indeed,

$$\lim_{x \rightarrow 0} \psi(x) = \lim_{r \rightarrow 0} |\log r|^s = \infty.$$

Therefore ψ is unbounded in every neighbourhood of the origin and hence

$$\psi \notin L^\infty(\Omega).$$

It remains to prove that $\psi \in W^{1,2}(\Omega)$.

For $x \neq 0$ we have by repeated chain rule,

$$\begin{aligned} \nabla \psi(x) &= s |\log |x||^{s-1} \frac{\log |x|}{|\log |x||} \frac{x}{|x|^2} \\ \implies |\nabla \psi(x)| &= s |\log |x||^{s-1} \left| \frac{\log |x|}{|\log |x||} \right| \frac{|x|}{|x|^2} \\ &= \frac{s |\log |x||^{s-1}}{|x|}. \end{aligned}$$

Moving again to polar coordinates,

$$\int_{\Omega} |\nabla \psi(x)|^2 dx = 2\pi s^2 \int_0^{1/2} \frac{|\log r|^{2s-2}}{r^2} r dr = 2\pi s^2 \int_0^{1/2} |\log r|^{2s-2} \frac{dr}{r}.$$

We substitute:

$$t = -\log r, \quad dt = -\frac{dr}{r},$$

we get

$$2\pi s^2 \int_{\log 2}^{\infty} t^{2s-2} dt.$$

The integral converges iff

$$2s - 2 < -1,$$

that is,

$$s < \frac{1}{2}.$$

Since this is precisely our assumption, we conclude that

$$\nabla \psi \in L^2(\Omega).$$

We have already shown that $\psi \in L^2(\Omega)$. Therefore

$$\psi \in W^{1,2}(\Omega).$$

Hence

$$\boxed{\psi \in W^{1,2}(\Omega), \quad \psi \in L^p(\Omega) \ \forall 1 \leq p < \infty, \quad \psi \notin L^\infty(\Omega).}$$

□

Example 1.3. $u(x) = \log \log \left(1 + \frac{1}{|x|}\right) \in W^{1,n}(B_1^n)$ for $n > 1$, but is unbounded near 0.

Proof. Since

$$1 + \frac{1}{|x|} \rightarrow \infty \quad \text{as } x \rightarrow 0,$$

we have

$$u(x) = \log \log \left(1 + \frac{1}{|x|}\right) \rightarrow \infty \quad \text{as } x \rightarrow 0.$$

Hence u is unbounded near 0.

We first show that $u \in L^n$. Moving to polar coordinates,

$$\int_{B_1^n} |u|^n dx = c_1 \int_0^1 \left[\log \log \left(1 + \frac{1}{r}\right) \right]^n r^{n-1} dr, \quad c_1 \in \mathbb{R}$$

Since $\log \log(1 + r^{-1})$ grows slower than any power of r^{-1} , the above integral converges. Therefore

$$u \in L^n(B_1^n).$$

We compute the gradient of u . Let

$$f(r) = \log \log \left(1 + \frac{1}{r}\right).$$

Then

$$f'(r) = \frac{1}{\log(1+r^{-1})} \cdot \frac{d}{dr} \log(1+r^{-1}) = -\frac{1}{r(r+1)\log(1+r^{-1})}.$$

Also,

$$\begin{aligned} \nabla u(x) &= f'(|x|) \frac{x}{|x|}, \\ |\nabla u(x)| &= \frac{1}{|x|(1+|x|)\log(1+|x|^{-1})}. \end{aligned}$$

Moving again to polar coordinates now,

$$\int_{B_1^n} |\nabla u|^n dx = c_2 \int_0^1 \frac{r^{n-1}}{r^n(1+r)^n[\log(1+r^{-1})]^n} dr, \quad c_2 \in \mathbb{R}.$$

Since $(1+r)^n \geq 1$,

$$\int_{B_1^n} |\nabla u|^n dx \leq C \int_0^1 \frac{dr}{r[\log(1+r^{-1})]^n}.$$

Since

$$\log(1+r^{-1}) \sim \log \frac{1}{r} \quad (r \rightarrow 0),$$

it suffices to study

$$\int_0^1 \frac{dr}{r(\log(1/r))^n}.$$

Using the substitution

$$t = \log \frac{1}{r}, \quad dt = -\frac{dr}{r},$$

we obtain

$$\int_0^1 \frac{dr}{r(\log(1/r))^n} = \int_{\log 1}^{\infty} t^{-n} dt.$$

which converges if and only if

$$n > 1.$$

Hence

$$\nabla u \in L^n(B_1^n).$$

Since $u \in L^n(B_1^n)$ and $\nabla u \in L^n(B_1^n)$, we conclude that

$$u \in W^{1,n}(B_1^n).$$

□

1.2. Theorems

Theorem 1.4 (Density of smooth functions up to the boundary). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth (also works for C^1) boundary and let*

$$1 \leq p < \infty.$$

Then

$$\overline{C^\infty(\overline{\Omega})}^{W^{1,p}(\Omega)} = W^{1,p}(\Omega).$$

In particular,

$$\overline{C^1(\overline{\Omega})}^{W^{1,p}(\Omega)} = W^{1,p}(\Omega).$$

Proof. For this purpose it suffices to show that $\overline{C^1(\overline{\Omega})}^{W^{1,p}(\Omega)} \subset W^{1,p}(\Omega)$. As the other inclusion is trivial.

Let

$$u \in W^{1,p}(\Omega).$$

We extend u to the whole space. Since $\partial\Omega$ is smooth, the extension theorem gives us a function

$$U := Pu \in W^{1,p}(\mathbb{R}^n)$$

such that

$$U|_\Omega = u \quad \text{and} \quad \|U\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\Omega)}.$$

Let ρ_ε be the usual mollifier and define

$$U_\varepsilon = \rho_\varepsilon * U.$$

Then

$$U_\varepsilon \in C^\infty(\mathbb{R}^n)$$

and, by the mollification theorem,

$$U_\varepsilon \rightarrow U \quad \text{in } W^{1,p}(\mathbb{R}^n).$$

We want to restrict back to Ω . So define,

$$u_\varepsilon := U_\varepsilon|_\Omega.$$

Since U_ε is smooth on all of $\mathbb{R}^n$, we have

$$u_\varepsilon \in C^\infty(\overline{\Omega}).$$

Moreover,

$$u_\varepsilon - u = (U_\varepsilon - U)|_\Omega.$$

Therefore

$$\|u_\varepsilon - u\|_{W^{1,p}(\Omega)} = \|(U_\varepsilon - U)|_\Omega\|_{W^{1,p}(\Omega)}.$$

Since restriction cannot increase the Sobolev norm,

$$\|(U_\varepsilon - U)|_\Omega\|_{W^{1,p}(\Omega)} \leq \|U_\varepsilon - U\|_{W^{1,p}(\mathbb{R}^n)}.$$

Hence

$$\|u_\varepsilon - u\|_{W^{1,p}(\Omega)} \leq \|U_\varepsilon - U\|_{W^{1,p}(\mathbb{R}^n)} \longrightarrow 0$$

as $\varepsilon \rightarrow 0$.

Thus

$$u_\varepsilon \rightarrow u \quad \text{in } W^{1,p}(\Omega) \quad \text{with } u_\varepsilon \in C^\infty(\bar{\Omega}).$$

We conclude that

$$\overline{C^\infty(\bar{\Omega})}^{W^{1,p}(\Omega)} \subset W^{1,p}(\Omega).$$

Since

$$C^\infty(\bar{\Omega}) \subset C^1(\bar{\Omega}),$$

we have

$$\overline{C^1(\bar{\Omega})}^{W^{1,p}(\Omega)} \subset W^{1,p}(\Omega).$$

□

Theorem 1.5. *Let $1 \leq p < \infty$. Then*

$$W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n).$$

Proof. The relation $W_0^{1,p} \subset W^{1,p}$ is easy to show. Since

$$W_0^{1,p}(\mathbb{R}^n) = \overline{C_c^\infty(\mathbb{R}^n)}^{W^{1,p}(\mathbb{R}^n)}$$

and

$$C_c^\infty(\mathbb{R}^n) \subset W^{1,p}(\mathbb{R}^n),$$

we have

$$W_0^{1,p}(\mathbb{R}^n) \subset W^{1,p}(\mathbb{R}^n).$$

The reverse inclusion requires a bit of work. Let

$$u \in W^{1,p}(\mathbb{R}^n).$$

Let $\eta \in C_c^\infty(\mathbb{R}^n)$ satisfy

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ on } B(0,1), \quad \eta \equiv 0 \text{ outside } B(0,2).$$

For $R > 0$, define

$$\eta_R(x) = \eta\left(\frac{x}{R}\right)$$

and let

$$u_R := \eta_R u.$$

So clearly,

$$u_R \in W^{1,p}(\mathbb{R}^n) \text{ and } \text{supp}(u_R) \subset B_{2R}(0).$$

Claim: $u_R \rightarrow u$ in $W^{1,p}(\mathbb{R}^n)$.

Notice

$$\begin{aligned} u_R - u &= (\eta_R - 1)u, \\ \implies |u_R - u|^p &\leq |u|^p. \end{aligned}$$

Also,

$$\eta_R(x) \rightarrow 1 \text{ for every } x \in \mathbb{R}^n.$$

Therefore, by DCT we have

$$\|u_R - u\|_{L^p(\mathbb{R}^n)} \rightarrow 0.$$

Next, we are interested in the convergence of the weak derivatives. For each $i = 1, \dots, n$, Leibniz' rule gives

$$D_i u_R = \eta_R D_i u + u D_i \eta_R, \quad \text{where } D_i \equiv \partial/\partial x_i$$

Hence

$$D_i u_R - D_i u = (\eta_R - 1)D_i u + u D_i \eta_R.$$

The first term converges to zero in $L^p(\mathbb{R}^n)$ by dominated convergence and owing to the fact that $D_i u \in L^p(\mathbb{R}^n)$.

For the second term,

$$D_i \eta_R(x) = \frac{1}{R} (D_i \eta) \left(\frac{x}{R} \right).$$

Thus

$$|D_i \eta_R| \leq \frac{C}{R}, \quad \text{as } \eta_R \in C_c^\infty(\mathbb{R}^n)$$

and

$$\text{supp}(D_i \eta_R) \subset A_R := \{R \leq |x| \leq 2R\}.$$

Therefore,

$$\|u D_i \eta_R\|_{L^p(\mathbb{R}^n)} \leq \frac{C}{R} \|u\|_{L^p(A_R)}.$$

Since

$$u \in L^p(\mathbb{R}^n),$$

we have

$$\|u\|_{L^p(A_R)} \rightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Therefore

$$\|u D_i \eta_R\|_{L^p(\mathbb{R}^n)} \rightarrow 0.$$

Hence

$$D_i u_R \rightarrow D_i u \quad \text{in } L^p(\mathbb{R}^n).$$

Combining the above convergences, we get

$$u_R \rightarrow u \quad \text{in } W^{1,p}(\mathbb{R}^n).$$

Let ρ_ε be the usual mollifier and define

$$u_{R,\varepsilon} = \rho_\varepsilon * u_R.$$

Since u_R has compact support,

$$u_{R,\varepsilon} \in C_c^\infty(\mathbb{R}^n).$$

Moreover,

$$u_{R,\varepsilon} \rightarrow u_R \quad \text{in } W^{1,p}(\mathbb{R}^n)$$

as $\varepsilon \rightarrow 0$.

This is true indeed as,

$$D_i(\rho_\varepsilon * u_R) = \rho_\varepsilon * (D_i u_R),$$

and since

$$u_R, D_i u_R \in L^p(\mathbb{R}^n),$$

the mollification theorem gives

$$\rho_\varepsilon * u_R \rightarrow u_R \text{ and } \rho_\varepsilon * (D_i u_R) \rightarrow D_i u_R \text{ in } L^p(\mathbb{R}^n).$$

Hence, upto a subsequence we have

$$u_{R,\varepsilon} \rightarrow u \quad \text{in } W^{1,p}(\mathbb{R}^n).$$

Since

$$u_{R,\varepsilon} \in C_c^\infty(\mathbb{R}^n),$$

it follows that

$$u \in W_0^{1,p}(\mathbb{R}^n).$$

Therefore

$$W^{1,p}(\mathbb{R}^n) \subset W_0^{1,p}(\mathbb{R}^n).$$

Combining this with the reverse inclusion gives

$$W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n).$$

□

Theorem 1.6 (Existence of Trace operator on the upper-half space). *Let $N \geq 2$. Then there exists a bounded linear operator*

$$\gamma_0 : H^1(\mathbb{R}_+^N) \rightarrow L^2(\mathbb{R}^{N-1})$$

such that

$$\gamma_0(u) = u|_{\mathbb{R}^{N-1}}$$

whenever

$$u \in H^1(\mathbb{R}_+^N) \cap C(\overline{\mathbb{R}_+^N}).$$

Proof. We start by defining the trace on C^1 functions. Let

$$u \in C_c^1(\overline{\mathbb{R}_+^N}).$$

Fix $x' \in \mathbb{R}^{N-1}$. Since u has compact support, $u(x', t) = 0$ for sufficiently large t . Applying the Fundamental Theorem of Calculus to $t \mapsto |u(x', t)|^2$, we obtain

$$\begin{aligned} |u(x', 0)|^2 &= - \int_0^\infty \frac{d}{dt} |u(x', t)|^2 dt, \\ &= -2 \int_0^\infty u(x', t) u_t(x', t) dt, \\ &\leq 2 \int_0^\infty |u(x', t)| |u_t(x', t)| dt. \end{aligned}$$

Applying Young's inequality

$$2ab \leq a^2 + b^2,$$

we obtain

$$|u(x', 0)|^2 \leq \int_0^\infty (|u(x', t)|^2 + |u_t(x', t)|^2) dt.$$

Integrating with respect to x' and using Fubini's theorem gives

$$\int_{\mathbb{R}^{N-1}} |u(x', 0)|^2 dx' \leq \int_{\mathbb{R}^{N-1}} \int_0^\infty (|u(x', t)|^2 + |u_t(x', t)|^2) dt dx'.$$

Therefore

$$\|u(\cdot, 0)\|_{L^2(\mathbb{R}^{N-1})}^2 \leq \|u\|_{L^2(\mathbb{R}_+^N)}^2 + \left\| \frac{\partial u}{\partial x_N} \right\|_{L^2(\mathbb{R}_+^N)}^2.$$

Since

$$\left\| \frac{\partial u}{\partial x_N} \right\|_{L^2(\mathbb{R}_+^N)} \leq \left(\sum_{i=1}^N \left\| \frac{\partial u}{\partial x_i} \right\|_{L^2(\mathbb{R}_+^N)}^2 \right)^{1/2}.$$

we conclude that

$$\|u(\cdot, 0)\|_{L^2(\mathbb{R}^{N-1})} \leq \|u\|_{H^1(\mathbb{R}_+^N)}.$$

Thus the map

$$u \mapsto u(\cdot, 0)$$

defines a bounded linear operator on $C_c^1(\overline{\mathbb{R}_+^N})$ and hence on $C_c^\infty(\overline{\mathbb{R}_+^N})$.

Since $C_c^\infty(\overline{\mathbb{R}_+^N})$ is dense in $H^1(\mathbb{R}_+^N)$, for every $u \in H^1(\mathbb{R}_+^N)$ there exists a sequence $u_m \in C_c^\infty(\overline{\mathbb{R}_+^N})$ such that $u_m \rightarrow u$ in $H^1(\mathbb{R}_+^N)$.

Applying the previous estimate to $u_m - u_n$ for $\gamma_0(u) = u|_{\mathbb{R}^{N-1}}$ we get

$$\|\gamma_0(u_m) - \gamma_0(u_n)\|_{L^2(\mathbb{R}^{N-1})} \leq \|u_m - u_n\|_{H^1(\mathbb{R}_+^N)}.$$

Hence

$$\{\gamma_0(u_m)\} \text{ is a Cauchy sequence in } L^2(\mathbb{R}^{N-1}).$$

Since $L^2(\mathbb{R}^{N-1})$ is complete, there exists $g \in L^2(\mathbb{R}^{N-1})$ such that $\gamma_0(u_m) \rightarrow g$ in $L^2(\mathbb{R}^{N-1})$.

We define

$$\gamma_0(u) := g.$$

We pass through the limit on the estimate to get,

$$\|\gamma_0(u)\|_{L^2(\mathbb{R}^{N-1})} \leq \|u\|_{H^1(\mathbb{R}_+^N)}.$$

Therefore

$$\gamma_0 : H^1(\mathbb{R}_+^N) \rightarrow L^2(\mathbb{R}^{N-1})$$

is a bounded linear operator.

Now let

$$u \in H^1(\mathbb{R}_+^N) \cap C(\overline{\mathbb{R}_+^N}).$$

Extend u to $\mathbb{R}^N$ by reflection. Let ρ_ε be the usual mollifier and let $\zeta_m \in C_c^\infty(\mathbb{R}^N)$ be cutoff functions satisfying

$$0 \leq \zeta_m \leq 1, \quad \zeta_m \equiv 1 \text{ on } B(0, m), \quad \text{supp}(\zeta_m) \subset B(0, 2m).$$

Choose $\varepsilon_m \downarrow 0$ and define

$$u_m = \zeta_m(\rho_{\varepsilon_m} * u).$$

Then

$$u_m \in C_c^\infty(\overline{\mathbb{R}_+^N}) \text{ and } u_m \rightarrow u \text{ in } H^1(\mathbb{R}_+^N).$$

Moreover, since u is continuous,

$$u_m \rightarrow u \text{ locally uniformly on } \overline{\mathbb{R}_+^N}.$$

Hence

$$u_m(\cdot, 0) \rightarrow u(\cdot, 0) \text{ pointwise on } \mathbb{R}^{N-1}.$$

On the other hand,

$$u_m(\cdot, 0) = \gamma_0(u_m) \rightarrow \gamma_0(u) \text{ in } L^2(\mathbb{R}^{N-1}).$$

Passing to a subsequence if necessary, we obtain

$$\gamma_0(u) = u(\cdot, 0) \text{ a.e. on } \mathbb{R}^{N-1}.$$

Therefore

$$\gamma_0(u) = u|_{\mathbb{R}^{N-1}},$$

and we are done. □

Chapter 2

Dirichlet Principle

The entire motivation for proceeding with this chapter is to look at partial differential equations in a new light. We often encounter difficult (possibly non-linear) PDEs in mathematics that do not admit an “easy” solution. So we intend to instead transform those PDEs into minimization problems where the PDEs itself become the “derivative” of the functionals to be minimized. The underlying PDEs for the ‘energy’ functionals are called The Euler-Lagrange Equations. We introduce the most well-known and arguably the simplest example of the same: The Dirichlet Principle.

Let

$$\Omega \subset \mathbb{R}^n$$

be an open, bounded, C^1 domain.

We define the functional

$$I[u] := \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx.$$

Let

$$u_0 \in H^1(\Omega)$$

be given.

We consider the minimization problem

$$m := \inf \left\{ I[u] : u \in u_0 + W_0^{1,2}(\Omega) \right\}.$$

Observe that $u - u_0 \in W_0^{1,2}(\Omega)$. Equivalently, $u - u_0 \in \ker(T_{\partial\Omega})$, where $T_{\partial\Omega}$ denotes the trace operator.

We shall address the following questions:

1. Existence of a minimizer.

2. Uniqueness of a minimizer.
3. Validity of the Euler–Lagrange equation.

Question 1: Existence of a minimizer

Is I bounded below and m finite?

Answer. Trivially, as the integrand is positive we have $m \geq 0$. Also, $I[u_0] < \infty$, since $u_0 \in H^1(\Omega)$.

Hence

$$m \in \mathbb{R}^+.$$

By the definition of infimum, there exists a minimizing sequence

$$\{u_s\}_{s=1}^\infty \subset u_0 + W_0^{1,2}(\Omega)$$

such that

$$I[u_s] \rightarrow m.$$

Passing to a tail of the sequence if necessary, we may assume

$$I[u_s] \leq m + 1 \quad \text{for all } s.$$

Therefore

$$\frac{1}{2} \int_{\Omega} |\nabla u_s|^2 dx \leq m + 1,$$

and hence

$$\|\nabla u_s\|_{L^2(\Omega)} \leq \sqrt{2(m+1)} \quad \text{for all } s.$$

To proceed further we will need an extremely crucial inequality called the Poincaré Inequality.

Proposition 2.1 (Poincaré Inequality). *Let*

$$\Omega \subset \mathbb{R}^n$$

be open, bounded, and of class C^1 .

Then there exists a constant

$$C = C(\Omega) > 0$$

such that

$$\|u\|_{L^2(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)} \quad \text{for every } u \in W_0^{1,2}(\Omega).$$

Proof. First let

$$\varphi \in C_c^\infty(\Omega).$$

Since Ω is bounded, there exists a cube $Q_d := (-d, d)^n$ containing Ω , i.e. $\Omega \subset Q_d$. Note that we are talking about a bounded set, and not a finite measure set.

Fix $x = (x_1, \dots, x_n) \in \Omega$.

For any index $i \in \{1, \dots, n\}$, by the Fundamental Theorem of Calculus,

$$\varphi(x_1, \dots, x_i, \dots, x_n) = \int_{-d}^{x_i} \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) ds.$$

Since

$$\varphi \equiv 0$$

outside a compact subset of Ω , we may equally write

$$\varphi(x) = \int_{-d}^d \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \chi_{(-d, x_i)}(s) ds.$$

Hence

$$|\varphi(x)| \leq \int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right| ds.$$

Applying the Cauchy-Schwarz inequality,

$$\begin{aligned} |\varphi(x)| &\leq \left(\int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right|^2 ds \right)^{1/2} \left(\int_{-d}^d 1^2 ds \right)^{1/2} \\ &= \sqrt{2d} \left(\int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right|^2 ds \right)^{1/2}. \end{aligned}$$

Therefore,

$$|\varphi(x)|^2 \leq 2d \int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right|^2 ds.$$

Integrating over Ω , we obtain

$$\begin{aligned} \|\varphi\|_{L^2(\Omega)}^2 &= \int_{\Omega} |\varphi(x)|^2 dx \\ &\leq 2d \int_{\Omega} \int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right|^2 ds dx. \end{aligned}$$

Since $\Omega \subset (-d, d)^n$, Fubini's theorem gives us

$$\begin{aligned}
\|\varphi\|_{L^2(\Omega)}^2 &\leq 2d \int_{-d}^d \cdots \int_{-d}^d \int_{-d}^d \left| \frac{\partial \varphi}{\partial x_i}(x_1, \dots, s, \dots, x_n) \right|^2 ds dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n \\
&= 2d \int_{Q_d} \left| \frac{\partial \varphi}{\partial x_i}(x) \right|^2 dx \text{ Since } Q_d = (-d, d)^n,
\end{aligned}$$

Hence

$$\|\varphi\|_{L^2(\Omega)}^2 \leq 2d \left\| \frac{\partial \varphi}{\partial x_i} \right\|_{L^2(Q_d)}^2.$$

Summing over $i = 1, \dots, n$, we get

$$\|\varphi\|_{L^2(\Omega)}^2 \leq 2d \|\nabla \varphi\|_{L^2(\Omega)}^2.$$

Therefore

$$\|\varphi\|_{L^2(\Omega)} \leq \sqrt{2d} \|\nabla \varphi\|_{L^2(\Omega)}.$$

Since $C_c^\infty(\Omega)$ is dense in $W_0^{1,2}(\Omega)$, there exists a sequence $\varphi_k \in C_c^\infty(\Omega)$ such that $\varphi_k \rightarrow u$ in $W_0^{1,2}(\Omega)$.

Applying the above estimate to each φ_k , we have

$$\|\varphi_k\|_{L^2(\Omega)} \leq \sqrt{2d} \|\nabla \varphi_k\|_{L^2(\Omega)}.$$

Passing to the limit as $k \rightarrow \infty$, we get

$$\|u\|_{L^2(\Omega)} \leq \sqrt{2d} \|\nabla u\|_{L^2(\Omega)}.$$

Hence the Poincaré inequality holds for every $u \in W_0^{1,2}(\Omega)$.

And we are done. □

By the Poincaré inequality,

$$\|u_s - u_0\|_{L^2(\Omega)} \leq C \|\nabla u_s - \nabla u_0\|_{L^2(\Omega)}.$$

Hence

$$\begin{aligned}
\|u_s - u_0\|_{W^{1,2}(\Omega)} &= \|u_s - u_0\|_{L^2(\Omega)} + \|\nabla u_s - \nabla u_0\|_{L^2(\Omega)} \\
&\leq (C + 1) \|\nabla u_s - \nabla u_0\|_{L^2(\Omega)}.
\end{aligned}$$

Using the triangle inequality,

$$\|\nabla u_s - \nabla u_0\|_{L^2(\Omega)} \leq \|\nabla u_s\|_{L^2(\Omega)} + \|\nabla u_0\|_{L^2(\Omega)}.$$

Therefore

$$\|u_s - u_0\|_{W^{1,2}(\Omega)} \leq (C + 1) \left(\|\nabla u_s\|_{L^2(\Omega)} + \|\nabla u_0\|_{L^2(\Omega)} \right).$$

Since

$$\|\nabla u_s\|_{L^2(\Omega)} \leq \sqrt{2(m+1)},$$

we get

$$\|u_s - u_0\|_{W^{1,2}(\Omega)} \leq (C + 1) \left(\sqrt{2(m+1)} + \|\nabla u_0\|_{L^2(\Omega)} \right).$$

Hence $\{u_s\}_{s=1}^\infty$ is bounded in $W^{1,2}(\Omega)$. Now, $W^{1,2}(\Omega)$ is a Hilbert space. In particular, $W^{1,2}(\Omega)$ is a separable reflexive Banach space.

Thus, by the Banach–Alaoglu theorem, every bounded sequence in a reflexive Banach space admits a weakly convergent subsequence.

Therefore there exist $u \in W^{1,2}(\Omega)$ and a subsequence $\{u_{s_k}\}$ such that

$$u_{s_k} \rightharpoonup u \quad \text{in } W^{1,2}(\Omega).$$

This is the infinite-dimensional analogue of the Bolzano–Weierstrass theorem.

Roughly speaking, $u_{s_k} \rightharpoonup u$ in $W^{1,2}(\Omega)$ means that

$$u_{s_k} \rightharpoonup u \quad \text{in } L^2(\Omega) \text{ and } \nabla u_{s_k} \rightharpoonup \nabla u \quad \text{in } L^2(\Omega; \mathbb{R}^n).$$

Recall that a sequence $f_n \subset L^2(\Omega)$ is said to converge weakly to $f \in L^2(\Omega)$, written $f_n \rightharpoonup f$ in $L^2(\Omega)$, if

$$\int_{\Omega} f_n g \, dx \longrightarrow \int_{\Omega} f g \, dx \quad \text{for every } g \in L^2(\Omega).$$

More generally, if $1 < p < \infty$, a sequence $f_n \subset L^p(\Omega)$ is said to converge weakly to $f \in L^p(\Omega)$, written $f_n \rightharpoonup f$ in $L^p(\Omega)$, if

$$\int_{\Omega} f_n g \, dx \longrightarrow \int_{\Omega} f g \, dx \quad \text{for every } g \in L^{p'}(\Omega),$$

where

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

We have obtained a subsequence

$$u_{s_k} \rightharpoonup u \quad \text{in } W^{1,2}(\Omega).$$

We now show that

$$u \in u_0 + W_0^{1,2}(\Omega).$$

Since

$$u_{s_k} \in u_0 + W_0^{1,2}(\Omega),$$

we have

$$u_{s_k} - u_0 \in W_0^{1,2}(\Omega) \quad \forall k.$$

Observe that

$$u_{s_k} - u_0 \rightharpoonup u - u_0 \quad \text{in } W^{1,2}(\Omega).$$

Since $W_0^{1,2}(\Omega)$ is a closed linear subspace of $W^{1,2}(\Omega)$, it is convex. A standard result from functional analysis states that every closed convex subset of a Banach space is weakly closed.

Therefore

$$u - u_0 \in W_0^{1,2}(\Omega).$$

Hence

$$u \in u_0 + W_0^{1,2}(\Omega).$$

Thus the weak limit falls in the admissible space.

It remains to show that u is a minimizer.

We have

$$\frac{1}{2} \int_{\Omega} |\nabla u|^2 dx \leq \liminf_{k \rightarrow \infty} \frac{1}{2} \int_{\Omega} |\nabla u_{s_k}|^2 dx.$$

Since

$$I[u_{s_k}] \rightarrow m,$$

we get

$$I[u] \leq \liminf_{k \rightarrow \infty} I[u_{s_k}] = m.$$

On the other hand,

$$m = \inf \{I[v] : v \in u_0 + W_0^{1,2}(\Omega)\}.$$

Since we have already shown u is admissible,

$$I[u] \geq m.$$

Combining the two inequalities,

$$I[u] = m.$$

Hence u is a minimizer.

We have therefore proved the following theorem.

Theorem 2.2. *Let*

$$\Omega \subset \mathbb{R}^n$$

be open, bounded, and of class C^1 .

Let

$$u_0 \in H^1(\Omega).$$

Consider the minimization problem

$$m = \inf \left\{ \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx : u \in u_0 + W_0^{1,2}(\Omega) \right\}.$$

Then there exists

$$u \in u_0 + W_0^{1,2}(\Omega)$$

such that

$$\frac{1}{2} \int_{\Omega} |\nabla u|^2 dx = m.$$

In other words, the minimum is attained.

Question 2: Uniqueness of the Minimizer

Suppose that

$$u_1, u_2 \in u_0 + W_0^{1,2}(\Omega)$$

are both minimizers.

Then

$$I[u_1] = I[u_2] = m.$$

Consider

$$w := \frac{u_1 + u_2}{2}.$$

Since $u_1 - u_0, u_2 - u_0 \in W_0^{1,2}(\Omega)$, and $W_0^{1,2}(\Omega)$ is a vector space,

$$w - u_0 = \frac{1}{2}(u_1 - u_0) + \frac{1}{2}(u_2 - u_0) \in W_0^{1,2}(\Omega).$$

Hence

$$w \in u_0 + W_0^{1,2}(\Omega).$$

Therefore

$$I[w] \geq m.$$

On the other hand,

$$\begin{aligned} I[w] &= \frac{1}{2} \int_{\Omega} \left| \frac{\nabla u_1 + \nabla u_2}{2} \right|^2 dx \\ &= \frac{1}{8} \int_{\Omega} |\nabla u_1 + \nabla u_2|^2 dx. \end{aligned}$$

Using $|a + b|^2 = |a|^2 + |b|^2 + 2a \cdot b$, we get

$$I[w] = \frac{1}{8} \int_{\Omega} (|\nabla u_1|^2 + |\nabla u_2|^2 + 2\nabla u_1 \cdot \nabla u_2) dx.$$

By the inequality $2a \cdot b \leq |a|^2 + |b|^2$, it follows that

$$|\nabla u_1 + \nabla u_2|^2 \leq 2|\nabla u_1|^2 + 2|\nabla u_2|^2.$$

Hence

$$\begin{aligned} I[w] &\leq \frac{1}{4} \int_{\Omega} (|\nabla u_1|^2 + |\nabla u_2|^2) dx \\ &= \frac{1}{2} (I[u_1] + I[u_2]) \\ &= m. \end{aligned}$$

Therefore

$$I[w] = m.$$

Since equality holds in $2a \cdot b \leq |a|^2 + |b|^2$, we must have $a = b$. And as a result we also must have,

$$\nabla u_1 = \nabla u_2 \quad \text{a.e. in } \Omega.$$

Hence

$$\nabla(u_1 - u_2) = 0.$$

Therefore

$$u_1 - u_2 = c \quad \text{for some constant } c.$$

But $u_1 - u_2 \in W_0^{1,2}(\Omega)$. And the trace of $u_1 - u_2$ vanishes on $\partial\Omega$, so

$$c = 0.$$

Thus

$$u_1 = u_2.$$

Hence the minimizer is unique. $\square$

Question 3: Euler–Lagrange Equation

Let u be the unique minimizer.

Take $\varphi \in C_c^\infty(\Omega)$ and define

$$g(t) := I[u + t\varphi].$$

Why did we take $\varphi \in C_c^\infty(\Omega)$? That was to ensure that $u + t\varphi$ remains in the admissible class. Now $g(t)$ is just a one-variable function and

$$g(t) \geq g(0) \quad \forall t \in \mathbb{R}.$$

Therefore

$$g'(0) = 0.$$

Now

$$\begin{aligned} g(t) &= \frac{1}{2} \int_{\Omega} |\nabla u + t \nabla \varphi|^2 dx \\ &= \frac{1}{2} \int_{\Omega} (|\nabla u|^2 + 2t \nabla u \cdot \nabla \varphi + t^2 |\nabla \varphi|^2) dx. \end{aligned}$$

Differentiating,

$$g'(t) = \int_{\Omega} \nabla u \cdot \nabla \varphi \, dx + t \int_{\Omega} |\nabla \varphi|^2 dx.$$

Hence

$$g'(0) = \int_{\Omega} \nabla u \cdot \nabla \varphi \, dx.$$

Since $g'(0) = 0$,

$$\boxed{\int_{\Omega} \nabla u \cdot \nabla \varphi \, dx = 0 \quad \forall \varphi \in C_c^\infty(\Omega).}$$

By the definition of weak derivatives,

$$\int_{\Omega} \nabla u \cdot \nabla \varphi \, dx = - \int_{\Omega} (\Delta u) \varphi \, dx.$$

Therefore

$$\int_{\Omega} (\Delta u) \varphi \, dx = 0 \quad \forall \varphi \in C_c^\infty(\Omega).$$

Hence, by the fundamental theorem of calculus of variations,

$$\boxed{\Delta u = 0 \quad \text{in } \Omega}$$

in the weak sense.

Thus the minimizer of the Dirichlet energy is a weak solution of Laplace's equation. $\square$

Next, we present and prove two results that were crucially used in the preceding discussion.

Proposition 2.3. *Let*

$$1 \leq p < \infty.$$

Suppose

$$f_n \rightarrow f \quad \text{strongly in } L^p(\Omega).$$

Show that

$$f_n \rightharpoonup f \quad \text{weakly in } L^p(\Omega).$$

Proof. Since

$$f_n \rightarrow f \quad \text{in } L^p(\Omega),$$

we have

$$\|f_n - f\|_{L^p(\Omega)} \longrightarrow 0.$$

To prove weak convergence, it suffices to show that

$$\int_{\Omega} f_n g \, dx \longrightarrow \int_{\Omega} f g \, dx \quad \text{for every } g \in L^{p'}(\Omega),$$

where

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

So let

$$g \in L^{p'}(\Omega)$$

be arbitrary. Then

$$\left| \int_{\Omega} f_n g \, dx - \int_{\Omega} f g \, dx \right| = \left| \int_{\Omega} (f_n - f) g \, dx \right|.$$

Applying Hölder's inequality, we get

$$\left| \int_{\Omega} (f_n - f)g \, dx \right| \leq \|f_n - f\|_{L^p(\Omega)} \|g\|_{L^{p'}(\Omega)}.$$

Since

$$\|f_n - f\|_{L^p(\Omega)} \rightarrow 0 \quad \text{and} \quad \|g\|_{L^{p'}(\Omega)} < \infty,$$

it follows that

$$\left| \int_{\Omega} (f_n - f)g \, dx \right| \rightarrow 0.$$

Therefore

$$\int_{\Omega} f_n g \, dx \longrightarrow \int_{\Omega} f g \, dx.$$

Since this holds for every $g \in L^{p'}(\Omega)$, we conclude that

$$f_n \rightharpoonup f \quad \text{in } L^p(\Omega).$$

Hence strong convergence in $L^p(\Omega)$ implies weak convergence in $L^p(\Omega)$. $\square$

Proposition 2.4. *Let X be a Banach space and let*

$$C \subset X$$

be a convex subset.

Show that if C is strongly closed, then C is weakly closed.

Proof. Let

$$x \in X \setminus C.$$

Since C is convex and norm closed, the Hahn–Banach Separation Theorem implies that there exist $f \in X^*$ and $\alpha \in \mathbb{R}$ such that

$$f(y) \leq \alpha \quad \forall y \in C,$$

while

$$f(x) > \alpha.$$

Consider the set

$$U = \{z \in X : f(z) > \alpha\}.$$

Since $f \in X^*$, it is continuous with respect to the weak topology on X . Therefore U is weakly open.

Moreover,

$$x \in U,$$

because

$$f(x) > \alpha.$$

On the other hand, for every $y \in C$, we have

$$f(y) \leq \alpha,$$

and hence

$$y \notin U.$$

Thus

$$U \cap C = \emptyset.$$

Therefore x possesses a weakly open neighbourhood which does not intersect C . It follows that x does not belong to the weak closure of C .

Since $x \in X \setminus C$ was arbitrary, we conclude that

$$\overline{C}^w \subset C.$$

The reverse inclusion $C \subset \overline{C}^w$ is obvious. Hence

$$\overline{C}^w = C.$$

Therefore C is weakly closed. □

Remark 2.5. Since $W_0^{1,2}(\Omega)$ is a closed linear subspace of $W^{1,2}(\Omega)$, it is convex. Therefore, by the above exercise, $W_0^{1,2}(\Omega)$ is weakly closed in $W^{1,2}(\Omega)$. Consequently, if

$$u_n - u_0 \in W_0^{1,2}(\Omega) \quad \text{for all } n$$

and

$$u_n \rightharpoonup u \quad \text{in } W^{1,2}(\Omega),$$

then

$$u - u_0 \in W_0^{1,2}(\Omega).$$